

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 50

Question Count: 5

Code of Conduct

I **M .Meghana(192525435)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *M. Meghana*

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1.Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical application. Analyze performance, security, and manageability, then justify the better choice.

Introduction

A hypervisor is virtualization software that creates and manages multiple virtual machines on a single physical server. It allows different operating systems to run simultaneously while sharing hardware resources. Hypervisors are mainly classified into Type-1 (Bare Metal) and Type-2 (Hosted). Hypervisor is essential for ensuring high performance, security, and availability in cloud data centre hosting mission-critical applications.

Comparison of Type-1 and Type-2 Hypervisors

Feature	Type-1 Hypervisor	Type-2 Hypervisor
Installation	Installed directly on physical hardware	Installed on top of a host operating system
Performance	High performance with minimal overhead	Lower performance due to host Operating System overhead
Security	More secure with fewer attack points	Less secure because host Operating System can be attacked
Resource Management	Efficient allocation of CPU, memory, and storage	Resources shared with host Operating System
Scalability	Suitable for large cloud environments	Suitable for personal or testing environments
Examples	Microsoft Hyper-V, Xen	Oracle VirtualBox, VMware Workstation

Analysis

Performance

- Type-1 hypervisors communicate directly with hardware, reducing latency and improving VM performance.
- Type-2 hypervisors depend on the host operating system, resulting in additional overhead and slower execution.

Security

- Type-1 hypervisors have a smaller attack surface because they do not require a host operating system.
- Type-2 hypervisors inherit vulnerabilities from the host OS, increasing security risks.

Manageability

- Type-1 supports centralized management, live migration, load balancing, clustering, and high availability.
- Type-2 provides limited management features and is mainly used for development or testing.

Justification

Mission-critical applications require continuous availability, high performance, and strong security. Type-1 hypervisors provide:

- Better resource utilization
- Higher reliability

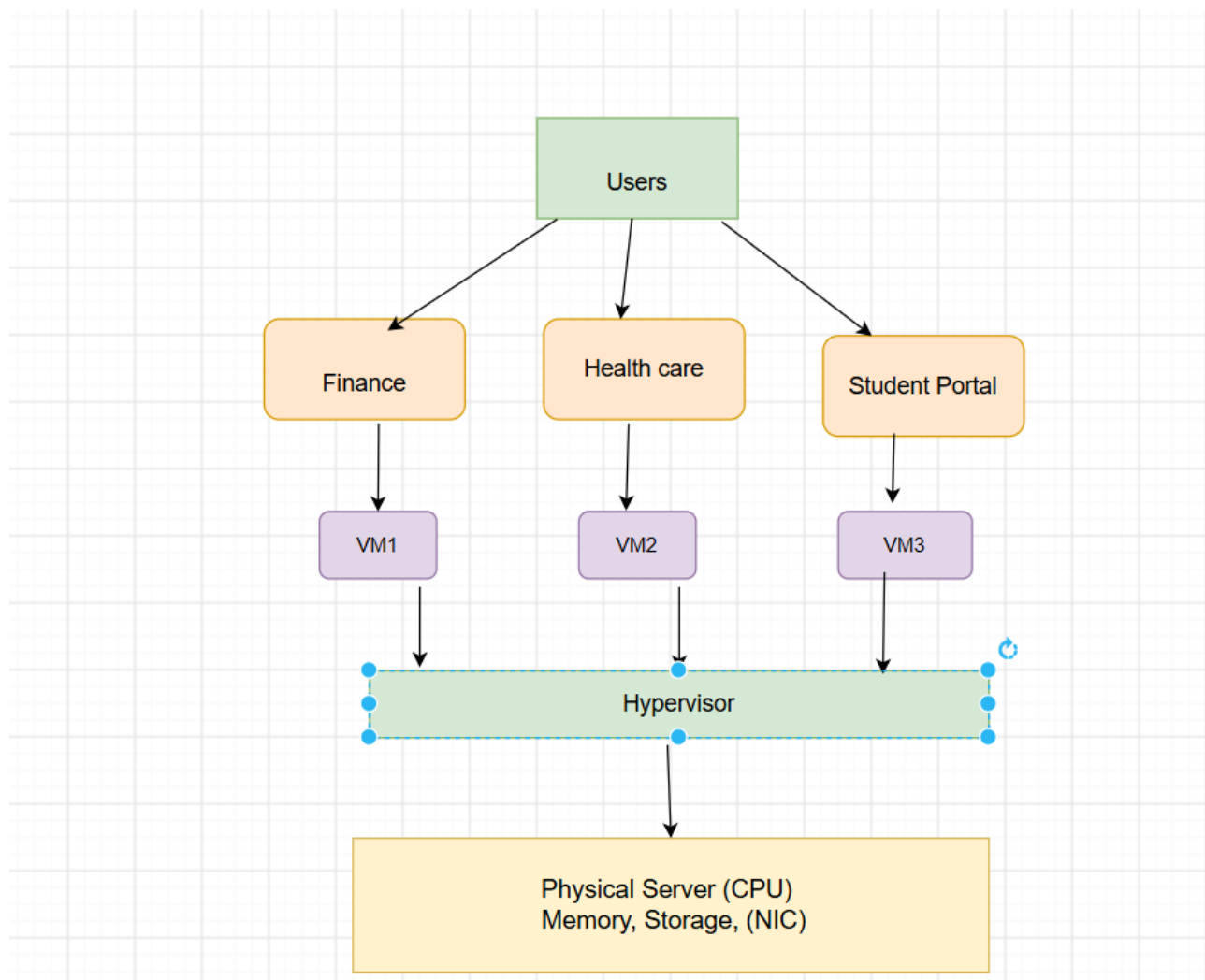
For cloud data centre hosting mission-critical applications, Type-1 Hypervisor is the preferred choice because it offers superior performance, stronger security, and advanced management capabilities.

2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Introduction

Virtualization enables multiple virtual machines to operate independently on the same physical server. Each VM runs its own operating system and applications while sharing hardware resources through a hypervisor. This architecture provides strong workload isolation, making it suitable for hosting multiple organizations or departments securely.

Architecture



How Virtualization Supports Isolation

- The hypervisor creates separate virtual Machine for finance, healthcare, and student portal applications.

- Each Virtual Machine has dedicated virtual CPU, memory, storage, and network resources.
- Failures or crashes in one Virtual Machine do not affect other Virtual Machine.
- Resource allocation is controlled to prevent interference between workloads.
- Virtual networking isolates communication among different applications.

Advantages

- Better resource utilization
- Improved security
- Independent operating systems
- Easy backup and recovery

Major Attack Surfaces

1. Hypervisor Attack

- Attackers exploit vulnerabilities in the hypervisor.
- A compromised hypervisor can expose all virtual machines hosted on the server.

2. Virtual Machine Escape Attack

- Malware escapes from one virtual machine and accesses the hypervisor Virtual Machine.
- This breaks isolation and compromises confidential data.

Security Measures

- Regularly update and patch the hypervisor.
- Use strong authentication and Role-Based Access Control (RBAC).
- Enable network segmentation using virtual LANs.
- Monitor Virtual Machine activities continuously.
- Encrypt sensitive data.

Virtualization architecture provides effective workload isolation and efficient resource sharing. However, securing the hypervisor and preventing VM escape attacks are essential for maintaining a secure cloud environment.

3.Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

Introduction

The utilization values indicate that the virtualization host is experiencing heavy resource usage, affecting virtual machine performance.

Analysis

CPU Utilization: 92%

- Indicates CPU saturation.
- Virtual machines compete for processing resources.

Corrective Action:

- Add more CPU cores.
- Migrate VMs to another host.
- Balance workloads.

Memory Utilization: 88%

- Indicates limited available RAM.
- May result in swapping and slower performance.

Corrective Action

- Increase physical memory.
- Optimize Virtual Machine memory allocation.

Storage I/O Latency: 35 ms

- High latency indicates slow storage access.
- Applications experience delays.

Corrective Action

- Upgrade to Software Defined Datacentre storage.
- Enable storage caching.
- Improve storage infrastructure.

VM Boot Time: 170 Seconds

- Slow startup caused by CPU, memory, and storage bottlenecks.

Corrective Action

- Optimize VM images.
- Reduce simultaneous VM startups.
- Improve storage performance.

Bottlenecks

- CPU overload
- Memory shortage
- Storage latency
- Resource contention

The host requires CPU, memory, and storage upgrades along with workload balancing to improve overall VM performance.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

Introduction

A Software Defined Data centre (SDDC) is a data centre where compute, storage, and network resources are virtualized and managed through software. Unlike a traditional data centre, which depends on physical hardware and manual configuration, SDDC provides automation, flexibility, and efficient resource management. It helps organizations quickly deploy and manage cloud services.

Traditional Data centre

In a traditional data centre, servers, storage, and network devices are managed separately. Resource provisioning is done manually, making deployment slow and increasing operational costs. Scaling the infrastructure also requires additional physical hardware.

Software Defined Data centre

1. Compute Virtualization

- Creates multiple virtual machines on a single physical server.
- Improves CPU and memory utilization.
- Enables quick deployment and migration of VMs.

2. Storage Virtualization

- Combines multiple storage devices into a single storage pool.
- Simplifies storage management.
- Supports backup and disaster recovery.

3. Network Virtualization

- Creates virtual networks through software.
- Improves network flexibility and security.
- Enables faster network configuration.

How SDDC Improves Agility

- Faster provisioning of resources.
- Automated infrastructure management.
- Easy scalability based on demand.
- Better utilization of hardware resources.
- Centralized management reduces administrative effort.

Comparison

Traditional Datacentre	SDDC
Manual management	Automated management
Slow deployment	Fast deployment
Limited scalability	Highly scalable
Hardware-based	Software-based

Software Defined Datacentre improves agility by virtualizing compute, storage, and networking resources. Compared to a traditional data centre, it provides faster deployment, better scalability, efficient resource utilization, and centralized management. Hence, SDDC is widely used in modern cloud computing environments.

5.A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Introduction

A multi-cloud federation allows organizations to use services from multiple cloud providers. However, different providers may use different identity management systems, leading to identity mismatches and authentication failures. Identity federation helps users access multiple cloud services securely using a single digital identity.

Issue Analysis

In a multi-cloud environment, each cloud provider may have its own authentication and authorization system. As a result:

- Users may need multiple login credentials.
- Authentication may fail due to identity mismatches.
- Access control policies may differ between providers.
- Managing user accounts becomes difficult and increases security risks.

Secure Identity Design

1. Identity Federation

- Establish trust between different cloud providers.
- Allow users to access multiple cloud services using one digital identity.

2. Single Sign-On (SSO)

- Users log in once and can access all authorized cloud services without logging in again.
- Improves user convenience and reduces password management.

3. Identity Provider (IdP)

- A trusted Identity Provider verifies user identities.
- Cloud providers rely on the IdP for authentication.

4. Privacy and Security Measures

- Encrypt user identity information during communication.
- Use Role-Based Access Control (RBAC) to restrict access.
- Enable Multi-Factor Authentication (MFA) for additional security.
- Maintain audit logs to monitor user activities.

Benefits:

- Secure authentication across multiple cloud providers.
- Improved user experience through Single Sign-On.
- Better protection of user privacy
- Consistent access control and improved security.

A secure multi-cloud federation can be achieved by using Identity Federation, Single Sign-On (SSO), Identity Provider RBAC, MFA, and encryption. These techniques eliminate identity mismatches, protect user privacy, and provide secure and seamless access across multiple cloud platforms.